

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (CE) Examination****November 2015****CE302/CE302.01 System Programming****Date: 28.11.2015, Saturday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections and both must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Write entire option in case of multiple choice and matching questions.

SECTION – I**Q-1 (A) Match the following.****[05]**

- | | |
|------------|---|
| 1) LEX | a) Macro |
| 2) #define | b) Text editor |
| 3) RDP | c) Language processing development tool |
| 4) TII | d) Parser |
| 5) Notepad | e) Single pass assembler |

(B) Answer the following question.**[06]**

- 1) Write regular expression (RE) of string over 0 and 1 ending with 0.
- 2) Define pattern and token.
- 3) What is symbol table? Which is the best data structure for implementation of symbol table.

Q-2 Answer the following questions. (Any three)**[12]**

- (A) Explain ORIGIN and EQU assembler directives.
- (B) Differentiate translator and interpreter.
- (C) What is assembly language? List and explain different type of assembly statements.
- (D) What do you mean by program relocation? What so you mean by self-relocatable program?

Q-3 Answer the following questions.

- (A) Explain data structures used in two pass assembler. **[05]**
- (B) Draw DFA for $1(0|1)^*0$ **[05]**
- (C) What is the use of LTORG statement? What happens if LTORG statement is not used in program? **[02]**

OR**Q-3 Answer the following questions.**

- (A) Explain (1) absolute loader (2) dynamic loader (3) relocation loader (4) bootstrap loader (5) linking loader **[05]**
- (B) What are the methods available for debugging in Turbo C++? **[05]**
- (C) Define editor. List types of editor. **[02]**

SECTION – II**Q-4(A) Choose the appropriate answer from given options.****[05]**

- 1) Which one of the following gap is between application domain and PL domain?

a) Semantic gap	b) Specification gap
c) Execution gap	d) None of above

- 2) Macro expansion is
 - a) Replace macro call with definition
 - b) Replace macro definition with call
 - c) Both (a) and (b)
 - d) None of above
- 3) Full name of CFL
 - a) Content free language
 - b) Content fix language
 - c) Context free language
 - d) Context fix language
- 4) Which one of the following RE will accept string 1010
 - a) (00)*
 - b) (11)*
 - c) (01)*
 - d) (10)*
- 5) Macro definition end with
 - a) MEND
 - b) MACRO END
 - c) Both (a) and (b)
 - d) None of above

(B) Answer the following questions.

[06]

- 1) Which are the facilities provided by debugger?
- 2) Define grammar.
- 3) What do you mean by conditional and semantic expansion of macro?

Q-5 Answer the following questions. (Any three)

[12]

- (A)** Write advantages of assembly language.
- (B)** What is nested macro? Explain how the nested macro call is processed? How stack is used to manage the data structures of macro in nested call?
- (C)** Explain java interpreter.
- (D)** List types of language processors with their input and outputs.

Q-6 Answer the following questions.

- (A)** What is preprocessor and macro processor? **[03]**
- (B)** What do you mean by left recursive grammar? Explain the method of removing left recursion from grammar with example. **[04]**
- (C)** Explain two pass macro processor. **[05]**

OR

Q-6 Answer the following questions.

- (A)** Differentiate subroutine and macro. **[03]**
- (B)** What do you mean by phase and pass of compiler? List phases of compiler. **[04]**
- (C)** What do you mean by ambiguous grammar? Explain with example. **[05]**
